

CPU Compare Module for Cortex-R4F (CCM-R4F)

This chapter describes the CPU compare module for Cortex-R4F (CCM-R4F). This device implements two instances of the Cortex-R4F CPU which are running in lock step to detect faults which may result in unsafe operating conditions. The CCM-R4F detects faults and signals them to an error signaling module (ESM).

NOTE: In general, R4F is used when referencing the Cortex CPU used in the Hercules family of devices; however, the floating-point functionality is a device-specific option and may not be included in some devices. Consult your device-specific datasheet to determine which core is included on your specific device being used.

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9.1 Main Features

Safety-critical applications require run-time detection of faults in the Central Processing Unit (CPU). For this purpose, the CPU Compare Module for Cortex-R4F (CCM-R4F) compares the core compare bus outputs of two Cortex-R4F CPUs running in a 1oo1D (one-out-of-one, with diagnostics) lockstep configuration. Any difference in the core compare bus outputs of the CPUs is flagged as an error. For diagnostic purposes, the CCM-R4F also incorporates a self-test capability to allow for boot time checking of hardware faults within the CCM-R4F itself.

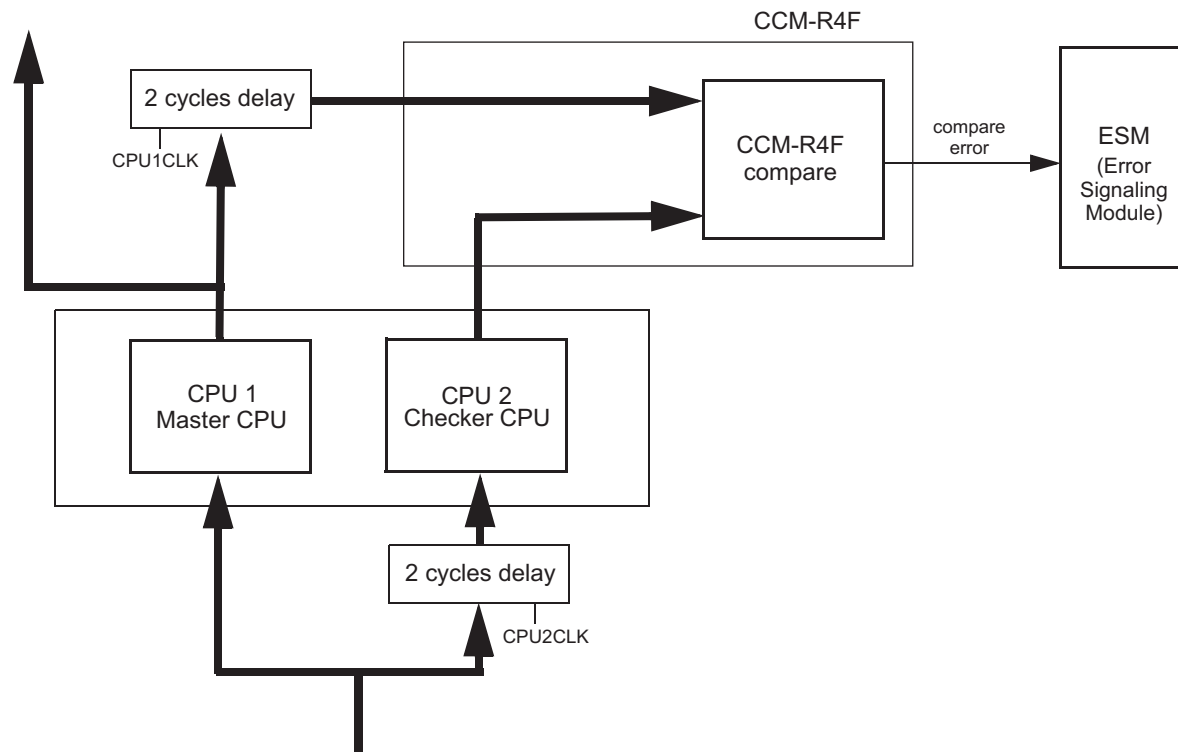
The main features of the CCM-R4F are:

- run-time detection of faults
- self-test capability
- error forcing capability

9.2 Block Diagram

Figure 9-1 shows the interconnection diagram of the CCM-R4F with the two Cortex-R4F CPUs. The core compare bus outputs of the CPUs are compared in the CCM-R4F. To avoid common mode impacts, the signals of the CPUs to be compared are temporally diverse. The output signals of the master CPU are delayed 2 cycles while the input signals of checker CPU are delayed 2 cycles. The CCM-R4F is constantly comparing about 900 signals from each of the two CPUs. These signals include the address, data and control signals from the flash and RAM TCMs and from the AXI peripheral bus. An internal register or ALU error will be flagged as a mis-compare when the faulty value is stored, used as an index, or causes a change in program execution.

Figure 9-1. Block Diagram



9.3 Module Operation

The CCM-R4F compares the core compare bus outputs of the master and checker Cortex-R4F CPUs on the microcontroller and signals an error on any mismatch. This comparison is started 6 CPU clock cycles after the CPU comes out of reset to ensure that CPU output signals have propagated to a known value after reset. Once comparison is started, the CCM module continues to monitor the outputs of two CPUs without any software intervention. Upon an error software needs to handle it.

The CCM-R4F can run in one of the following four operating modes:

1. 1001D lock step
2. self-test
3. error forcing
4. self-test error forcing

The operating mode can be selected by writing a dedicated key to the key register (MKEY).

9.3.1 1001D Lock Step Mode

This is the default mode on start-up.

In lock step mode, the compare bus output signals of both CPUs are compared. A difference in the CPU compare bus outputs is indicated by signaling an error to the ESM which sets the error flag “CCM-R4F - compare”.

NOTE: The CPU compare error asserts “CCM-R4F self-test error” flag as well. By doing this, the CPU compare error has two paths (“CCM-R4F - compare” and “CCM-R4F self-test error” flag) to the ESM, so that even if one of the paths fails, the error is still propagated to the ESM.

Not all internal registers of the Cortex-R4F CPU have fixed values upon reset. To avoid an erroneous CCMR4F compare error, the application software needs to ensure that the CPU registers of both CPUs are initialized with the same values before the registers are used, including function calls where the register values are pushed onto the stack.

9.3.2 Self-Test Mode

In self-test mode, the CCM-R4F checks itself for faults. During self-test, the compare error module output signal is deactivated. Any fault detected inside the CCM-R4F will be flagged by ESM error “CCM-R4F - self-test”.

In self-test mode, the CCM-R4F automatically generates test patterns to look for any hardware faults. If a fault is detected, then a self-test error flag is set, a self-test error signal is asserted and sent to the ESM, and the self-test is terminated immediately. If no fault is found during self-test, the self-test complete flag is set. In both cases, the CCM-R4F remains in self-test mode after the test has been terminated or completed, and the application needs to switch the CCM-R4F mode by writing another key to the mode key register (MKEY). During the self-test operation, the compare error signal output to the ESM is inactive irrespective of the compare result.

There are two types of patterns generated by CCM-R4F during self-test mode:

- i. Compare Match Test
- ii. Compare Mismatch Test

CCM-R4F first generates Compare Match Test patterns, followed by Compare Mismatch Test patterns. Each test pattern is applied on both CPU signal inputs of the CCM-R4F's compare block and clocked for one cycle. The duration of self-test is 3615 CPU clock cycles (GCLK).

NOTE: During self-test, both CPUs can execute normally, but the compare logic will not be checking any CPU signals. Also during self-test, only the compare unit logic is tested and not the memory mapped register controls for the CCM-R4F. The self-test is not interruptible.

9.3.2.1 Compare Match Test

During the Compare Match Test, there are four different test patterns generated to stimulate the CCM-R4F. An identical vector is applied to both input ports at the same time expecting a compare match. These patterns cause the self-test logic to exercise every CPU compare bus output signal in parallel. If the compare unit produces a compare mismatch then the self-test error flag is set, the self-test error signal is generated, and the Compare Match Test is terminated.

The four test patterns used for the Compare Match Test are:

- All 1s on both CPU signal ports
- All 0s on both CPU signal ports
- 0xAs on both CPU signal ports
- 0x5s on both CPU signal ports

These four test patterns will take four clock cycles to complete. illustrates the sequence of Compare Match Test.

Table 9-1. Compare Match Test Sequence

CPU 1 Signal Position									CPU 2 Signal Position									Cycle
n:8	7	6	5	4	3	2	1	0	n:8	7	6	5	4	3	2	1	0	
1s	1	1	1	1	1	1	1	1	1s	1	1	1	1	1	1	1	1	0
0s	0	0	0	0	0	0	0	0	0s	0	0	0	0	0	0	0	0	1
0xA	1	0	1	0	1	0	1	0	0xA	1	0	1	0	1	0	1	0	2
0x5	0	1	0	1	0	1	0	1	0x5	0	1	0	1	0	1	0	1	3

9.3.2.2 Compare Mismatch Test

During the Compare Mismatch Test, the number of test patterns is equal to twice the number of CPU output signals to compare in lock step mode. An all 1s vector is applied to the CCM-R4F's CPU1 input port and the same pattern is also applied to the CCM-R4F's CPU2 input port but with one bit flipped starting from signal position 0. The un-equal vector will cause the CCM-R4F to expect a compare mismatch at signal position 0, if the CCM-R4F logic is working correctly. If, however, the CCM-R4F logic reports a compare match, the self-test error flag is set, the self-test error signal is asserted, and the Compare Mismatch Test is terminated.

This Compare Mismatch Test algorithm repeats in a domino fashion with the next signal position flipped while forcing all other signals to logic level 1. This sequence is repeated until every single signal position is verified on both CPU signal ports.

The Compare Mismatch Test is terminated if the CCM-R4F reports a compare match versus the expected compare mismatch. This test ensures that the compare unit is able to detect a mismatch on every CPU signal being compared. [Table 9-2](#) illustrates the sequence of Compare Mismatch Test. There is no error signal is sent to ESM if the expected errors are seen with each pattern.

Table 9-2. Compare Mismatch Test Sequence

CPU 1 Signal Position										CPU 2 Signal Position										Cycle		
n	n-1:8		7	6	5	4	3	2	1	0	n	n-1:8		7	6	5	4	3	2		1	0
1	1	1s	1	1	1	1	1	1	1	1	1	1	1s	1	1	1	1	1	1	1	0	0
1	1	1s	1	1	1	1	1	1	1	1	1	1	1s	1	1	1	1	1	1	0	1	1
1	1	1s	1	1	1	1	1	1	1	1	1	1	1s	1	1	1	1	1	0	1	1	2
1	1	1s	1	1	1	1	1	1	1	1	1	1	1s	1	1	1	1	0	1	1	1	3
::																						
1	1	1s	1	1	1	1	1	1	1	1	1	0	1s	1	1	1	1	1	1	1	1	n-1
1	1	1s	1	1	1	1	1	1	1	1	0	1	1s	1	1	1	1	1	1	1	1	n
1	1	1s	1	1	1	1	1	1	1	0	1	1	1s	1	1	1	1	1	1	1	1	n+1
1	1	1s	1	1	1	1	1	1	0	1	1	1	1s	1	1	1	1	1	1	1	1	n+2
1	1	1s	1	1	1	1	1	0	1	1	1	1	1s	1	1	1	1	1	1	1	1	n+3
1	1	1s	1	1	1	1	0	1	1	1	1	1	1s	1	1	1	1	1	1	1	1	n+4
::																						
1	0	1s	1	1	1	1	1	1	1	1	1	1	1s	1	1	1	1	1	1	1	1	2n-1
0	1	1s	1	1	1	1	1	1	1	1	1	1	1s	1	1	1	1	1	1	1	1	2n

9.3.3 Error Forcing Mode

In error forcing mode, a test pattern is applied to the CPU related inputs of the CCM-R4F compare logic to force an error in the compare error output signal of the compare unit. The ESM error flag “CCM-R4F - compare” is expected after the error forcing mode completes. As a side effect, the “CCM-R4F self-test error” flag is also asserted whenever the CPU compare error is asserted.

Error forcing mode is similar to the Compare Mismatch Test operation of self-test mode in which an unequal vector is applied to the CCM-R4F CPU signal ports. The error forcing mode forces the compare mismatch to actually assert the compare error output signal. This ensures that faults in the path between CCM-R4F and ESM is detected.

Only one hardcoded test pattern is applied into CCM-R4F during error forcing mode. A repeated 0x5 pattern is applied to CPU1 signal port of CCM-R4F input while a repeated 0xA pattern is applied to the CPU2 signal port of CCM-R4F input. The error forcing mode takes one cycle to complete. Hence, the failing signature is presented for one clock cycle. After that, the mode is automatically switched to lock step mode. The key register (MKEY) will indicate the lock step key mode once it is switched to lock step mode. During the one cycle required by the error forcing test, the CPU output signals are not compared. User should expect the ESM to trigger a response (report the CCM-R4F fail). If no error is detected by ESM, then a hardware fault is present.

9.3.4 Self-Test Error Forcing Mode

In self-test error forcing mode, an error is forced at the self-test error signal. The compare unit is still running in lockstep mode and the key is switched to lockstep after one clock cycle. The ESM error flag “CCM-R4F - self-test” is expected after the self-test error forcing mode completes. Once the expected errors are seen, the application can clean the error through ESM module.

9.3.5 Operation During CPU Debug Mode

Certain debug operations place the CPU in a halting debug state where the code execution is halted. Because halting debug events are asynchronous, there is a possibility for the debug requests to cause loss of lockstep. CCM-R4F will disable upon detection of halting debug requests. Core compare error will not be generated and flags will not update. A CPU reset is needed to ensure the CPUs are again in lockstep and will also re-enable the CCM-R4F.

9.4 CCM-R4F Control Registers

[Table 9-3](#) lists the CCM-R4F registers. Each register begins on a 32-bit word boundary. The registers support 32-bit, 16-bit and 8-bit accesses. The base address for the control registers is FFFF F600h.

Table 9-3. CCM-R4F Control Registers

Offset	Acronym	Register Description	Section
00h	CCMSR	CCM-R4F Status Register	Section 9.4.1
04h	CCMKEYR	CCM-R4F Key Register	Section 9.4.2

9.4.1 CCM-R4F Status Register (CCMSR)

Figure 9-2. CCM-R4F Status Register (CCMSR) (Address = FFFF F600h)

31											17	16		
Reserved											CPME			
R-0											R/WPC-0			
15								9	8	7	2	1	0	
Reserved							STC	Reserved				STET	STE	
R-0							R-0		R-0				R-0	R-0

LEGEND: R/W = Read/Write; R = Read only; C = Clear; WP = Write in Privileged mode only; -n = value after reset

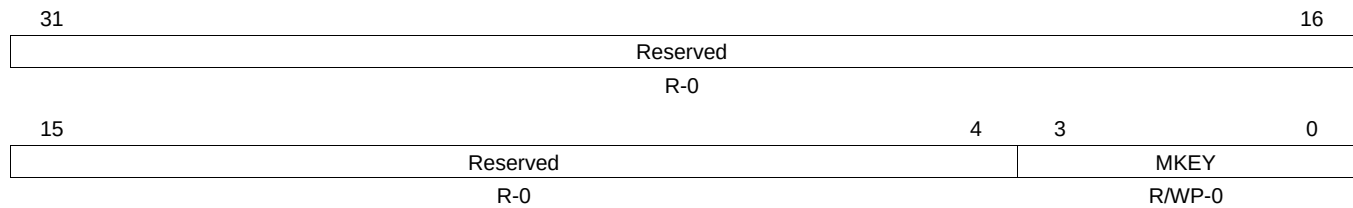
Table 9-4. CCM-R4F Status Register (CCMSR) Field Descriptions ⁽¹⁾

Bit	Field	Value	Description
31-17	Reserved	0	Reads return zeros and writes have no effect.
16	CPME	0	Compare Error Read in User and Privileged mode. Write in Privileged mode only. Read: CPU signals are identical. Write: Leaves the bit unchanged.
		1	Read: CPU signal compare mismatch. Write: Clears the bit.
15-9	Reserved	0	Reads return zeros and writes have no effect.
8	STC	0	Self-test Complete Note: This bit is always 0 when not in self-test mode. Once set, switching from self-test mode to other modes will clear this bit. Read/Write in User and Privileged mode. Read: Self-test on-going if self-test mode is entered. Write: Writes have no effect.
		1	Read: Self-test is complete. Write: Writes have no effect.
7-2	Reserved	0	Reads return zeros and writes have no effect.
1	STET	0	Self-test Error Type Read/Write in User and Privileged mode. Read: Self-test failed during Compare Match Test if STE = 1. Write: Writes have no effect.
		1	Read: Self-test failed during Compare Mismatch Test if STE = 1. Write: Writes have no effect.
0	STE	0	Self-test Error Note: This bit gets updated when the self-test is complete or an error is detected. Read/Write in User and Privileged mode. Read: Self-test passed. Write: Writes have no effect.
		1	Read: Self-test failed. Write: Writes have no effect.

⁽¹⁾ The contents of this register should be interpreted in context of what test was selected. That is what mode is CCM operating in.

9.4.2 CCM-R4F Key Register (CCMKEYR)

Figure 9-3. CCM-R4F Key Register (CCMKEYR) (Address = FFFF F604h)



LEGEND: R/W = Read/Write; R = Read only; WP = Write in Privileged mode only; -n = value after reset

Table 9-5. CCM-R4F Key Register (CCMKEYR) Field Descriptions

Bit	Field	Value	Description
31-4	Reserved	0	Reads return to zeros and writes have no effect.
3-0	MKEY		Mode Key
		0	Read in User and Privileged mode. Write in Privileged mode only. Read: Returns current value of the MKEY. Write: Lockstep mode.
		6h	Read: Returns current value of the MKEY. Write: Self-test mode.
		9h	Read: Returns current value of the MKEY. Write: Error Forcing mode.
		Fh	Read: Returns current value of the MKEY. Write: Self-test Error Forcing mode.
		Other values	Note: It is recommended to not write any other key combinations. Invalid keys will result in switching operation to lockstep mode.